

MISO OMNISEARCH: PREDICTIVE & COMPARATIVE KNOWLEDGE ENGINE

Session-Aware Pre-Fetching, 360° Knowledge Canvas, Side-by-Side Comparison & 1-Click PDF Studio

Client / Challenge: MISO Prompt 1 (Accessibility, Discoverability, Usability)	Target Users: Power Traders, Municipal Utilities / Co-ops, Media & Regulators
Core Metaphor: Intelligent Discovery Engine (Perplexity + Bloomberg for MISO)	Primary Goal: Deflect routine CSR data requests & empower proactive self-service

1. Executive Vision: Session-Aware Pre-Fetching & 360° Discovery

MISO OmniSearch is an intelligent discovery and comparison engine designed to solve MISO's information accessibility and discoverability challenges. Instead of making users manually search through nested web pages or guess what data exists, OmniSearch actively assists the user at every step:

- **Session-Aware Journey Pre-Fetching:** Silently observes the user's active browsing path on misoenergy.org (e.g. visiting Indiana Hub LMP reports → filtering the Solar Queue → viewing MTEP24). When the user opens OmniSearch, it instantly pre-loads a synthesized session briefing with isolated KPIs, relevant transmission upgrade stats, and suggested next actions before they type a single letter.
- **The 360° Knowledge Canvas:** Searching any query dynamically builds a structured visual workspace containing direct grounded answers, isolated KPI cards, live interactive Recharts curves, and side-by-side comparative matrices.

2. The Five Pillars of MISO OmniSearch

Pillar / Module	Core Capabilities & User Experience	Impact on Discoverability & Usability
1. Session-Aware Intent Pre-Fetching	• Tracks in-session context on misoenergy.org. • Pre-loads active state/hub statistics, recent queue filings, and relevant MTEP summaries. • Displays ready-to-click quick action pills on launch.	Zero Friction: Eliminates cold-start blank screens by anticipating what data the user is researching.
2. Unified OmniSearch & Auto-Suggest	• Single omni-input bar supporting plain-English questions, market hubs, acronyms, and report names. • Real-time autocomplete suggestions with instant category badges (Market Pricing, Fuel Mix, Transmission, Glossary).	Cross-Silo Search: Users find buried data across PDFs, CSVs, and APIs without knowing file paths.
3. 360° Knowledge Canvas & KPI Cards	• Direct Answer: Verified plain-English summary with exact PDF page citations. • KPI Cards: Isolated metrics (e.g. Real-Time Avg, Day-Ahead Spread, Peak Hour). • Interactive Recharts: 24-hr LMP area graphs and fuel mix donut charts.	Instant Synthesis: Delivers answers, numbers, and visualizations on a single screen without clicking through nested pages.
4. Side-by-Side Comparison Engine	• Compare Hubs: Indiana vs. Michigan vs. Texas (LMPs, congestion, losses). • Compare Fuel Types: Wind vs. Solar vs. Gas (GW capacity, records, growth). • Compare Plans: Local MTEP vs. Regional LRTP vs. Interregional JTIQ.	Deep Exploration: Point-and-click comparative matrices with synchronized multi-line Recharts curves.
5. Proactive Radar & 1-Click PDF Studio	• Proactive Next Questions: Clickable smart pills predicting the user's next logical inquiries. • 1-Click PDF Fact Sheet: Compiles the active canvas findings into a branded 1-page PDF in 3 seconds. • 1-Click filtered CSV dataset downloads.	Actionable Output: Fulfills custom briefing requests instantly and deflects repetitive CSR inquiries.

3. 5-Person Team Work Allocation Model

Role / Member	Core Focus & Responsibilities	OmniSearch Deliverables
Member 1 <i>Lead AI & Backend Engineer</i>	• Build FastAPI search pipeline & Gemini Flash recommendation engine • Session-context evaluation endpoint (`/api/session-prefetch`) • Automated ReportLab PDF generator service for 1-click fact sheets	• FastAPI search API • Session pre-fetch service • PDF generator engine
Member 2 <i>Frontend & Interactive Viz Lead</i>	• Next.js / React application with Tailwind CSS & responsive UI • Build 360° Knowledge Canvas, search auto-suggest, and comparison cards • Interactive Recharts comparison graphs & dynamic follow-up pills	• Live interactive search portal • Recharts comparison charts • 360° Canvas components
Member 3 <i>Data & Knowledge Graph Lead</i>	• Ingest MISO market datasets, fuel mix records, and MTEP data tables • Construct the 'Related Question & Comparison Mapping Graph' • Ingest and clean Legacy-to-API Crosswalk JSON and Jargon Glossary	• Knowledge Graph JSON • Curated Market Datasets • Crosswalk & Glossary store
Member 4 <i>UI/UX Designer & Accessibility Lead</i>	• Design the 360° Knowledge Canvas and comparison cards in Figma • WCAG 2.1 AA accessibility audit (screen reader, high contrast, keyboard nav) • Interactive Jargon Tooltips, 'X-Ray Mode', and audio voice briefing UI	• Figma interactive prototype • WCAG 2.1 AA Audit Report • Audio narration integration
Member 5 <i>Product Manager & Pitch Lead</i>	• Problem framing: how unified discovery eliminates 80%+ of routine CSR requests • Build financial & operational ROI model for Help Center ticket deflection • 10–12 slide presentation deck & live pitch script; direct 2-minute demo video	• 10–12 slide pitch deck • Quantified ROI model • 2-minute live demo script

4. The 2-Minute Live Showcase Demo Flow (How to Win)

Minute 0:00 – 0:30 (The Problem & The Hook): <i>'External partners spend hours hunting for data across MISO's silos. We built MISO OmniSearch: a predictive, session-aware intelligence engine.'</i>
Minute 0:30 – 1:00 (Session-Aware Pre-Fetching): Open OmniSearch → Highlight how it instantly recognizes the user was viewing Indiana Hub and pre-loads the 360° Knowledge Canvas with live KPIs without typing a single letter.
Minute 1:00 – 1:30 (The Comparison Engine & Proactive Radar): Click <i>'Compare with Michigan Hub'</i> → Chart overlays both curves and reveals \$12 congestion spread → Click proactive pill <i>'Show Wind vs Gas'</i> to seamlessly pivot data.
Minute 1:30 – 2:00 (1-Click PDF Export & ROI): Click <i>'Export Fact Sheet PDF'</i> to download a branded 1-page report → Conclude with 80% reduction in routine CSR data requests.

5. Immediate Action Plan

- 1. Send PDF to Team:** Share this clean 2-page blueprint for final sign-off.
- 2. Build Frontend OmniSearch Canvas:** Wire the OmniSearch input bar, session pre-fetch banner, comparison matrix, and proactive pills into the React frontend.
- 3. Build Backend Comparison APIs:** Implement `/api/compare` (Hubs, Fuels, Transmission) and ReportLab 1-click PDF briefing exporter.